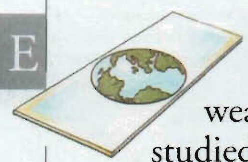
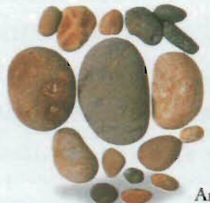
EARTH SCIENCES



FOSSILS PROVIDE CLUES to the ages of rocks; the atmosphere provides clues to tomorrow's weather. Amongst others, these elements are studied within the discipline of Earth sciences. This is the study of the planet's physical characteristics, from volcanoes to raindrops. The different branches of Earth sciences cover all of the Earth's dynamic systems, apart from life forms, which are studied within biology. Knowledge about the Earth's history and formation also informs us about its needs, which will help ensure the future survival of the planet.

Branches of Earth sciences

The term Earth sciences has been used since the 1970s. It covers the range of subjects that were previously bracketed under the term "physical geography". Although each of the Earth sciences is a distinct study focusing on one aspect of the Earth, each is also a key element of the inter-related study of Earth sciences.



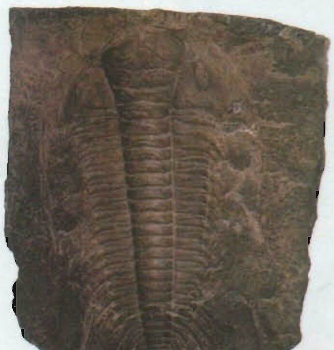
Pebbles

Anthracite, a form of coal

Granite

Geology

The oldest branch of the Earth sciences, geology is the study of the Earth's history, structure, and make-up. Although it centres on rocks and the composition of the Earth's crust, geology also relates to the other Earth sciences, except for meteorology.



Fossil of a sea creature

Palaeontology

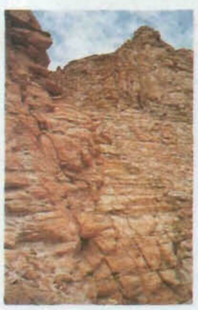
Fossils, the remains of once living organisms preserved in sedimentary rock, are studied within the branch of Earth science called palaeontology. From fossils, scientists can work out the ages of rocks and develop a picture of the history of plant and animal life on Earth over billions of years.

Earth sciences cover many different areas of study.

Volcanology

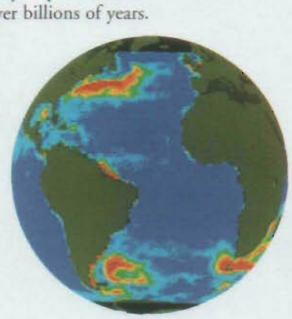
The study of volcanoes, and the reasons why they erupt, is known as volcanology. It may involve volcanologists working close to an erupting volcano. The scientists wear special clothing to protect them from gas, heat, and flying lava bombs.

Volcanic bombs



Geomorphology

The study of landforms and the processes that shape them is known as geomorphology. It includes landforms ranging from mountains and valleys to rivers and glaciers, and the effects of different shaping processes upon them, such as the erosion caused by weathering.



Oceanography

The study of the oceans is called oceanography. It covers ocean chemistry, the ocean bed and currents (shown above by satellite), and marine life.

Geography

This is the study of the Earth's surface. Human geography looks at world patterns of human activity; physical geography studies the Earth's physical environment.

Meteorology

The atmosphere is studied within the discipline of meteorology. This focuses on the processes that make the weather, and on weather forecasting. Climatology is the study of weather patterns.

Surveying the Earth

Earth scientists can learn very little about the Earth from laboratory studies. Instead, they must make observations, collect data, and test their theories in the outside world – this may mean climbing mountains or braving earthquakes. Satellite photography has provided a vast new source of data, but most information continues to come from field work.

Survey equipment

Earth scientists sometimes need to use specific survey equipment. This laser equipment helps to monitor the movement of earthquakes.

Earth resources

The Earth provides all the materials we need for living, from the food we eat and the water we drink, to the bricks we use for building. Earth sciences help us to identify the location of these resources. They also show what damage we may be doing to them by exploiting them thoughtlessly.

Fruit

Squid

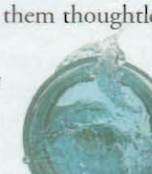
Air

We need air to breathe virtually every second of our lives. However, this vital resource is becoming increasingly damaged by human pollution.

Tourmaline gemstone

Minerals

From metal for cars to concrete for buildings, nearly everything we make comes from the minerals or chemicals taken from the Earth's crust. Gems are another of its rich resources.



Water

All forms of life are dependent on water. Patterns of human activity are controlled by the need to be near a source of clean water.

Food

Food is provided by things living on the Earth's surface. These depend on the mineral resources, water, and air provided by the Earth.

Energy

Ninety per cent of the energy we use comes from a finite supply of minerals – oil, coal, and gas – extracted from the Earth's crust.

FIND OUT MORE

CLIMATE · EARTH · FOSSILS · GEOLOGY · OCEANS AND SEAS · ROCKS AND MINERALS · VOLCANOES · WEATHER